

Tokenizing Patents as Participatory Engines: Blockchain-Driven Democratization of Innovation Ecosystems

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This study demonstrates how blockchain-based tokenization of patents using ERC-20 standards transcends financial efficiency to reshape social access to innovation ecosystems. Through large-scale simulations of 20 million transactions, we forecast the emergence of dynamic intellectual property (IP) markets capable of democratizing asset ownership, reducing historical inequalities in IP access, and fostering global collaborative innovation. Our findings reveal that tokenization significantly enhances liquidity, particularly in high-innovation sectors like technology and healthcare, by transforming traditionally illiquid patents into tradable digital assets. Crucially, liquidity gains are moderated by sectoral disparities, regulatory frameworks, and token design, suggesting risks of a "tokenization divide." Machine learning-driven analysis further predicts how liquidity dynamics could evolve under varying adoption scenarios. We propose actionable pathways for policymakers to harness this disruption: adaptive governance models that promote inclusive participation, anti-concentration mechanisms, and cross-border compliance standards. By bridging empirical validation with socio-technical forecasting, this work advances a future where tokenization redefines IP not merely as an asset class, but as a participatory engine for equitable innovation.

Keywords: Blockchain tokenization, Intellectual property markets, Innovation democratization, Liquidity enhancement, Decentralized finance (DeFi), Innovation ecosystems

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1. Introduction

The global intellectual property (IP) market, valued at over \$1.7 trillion, stands at an inflection point. Traditional IP management, characterized by opacity, illiquidity, and institutional gatekeeping (Lemley, 2004; Fairfield, 2015), perpetuates stark inequalities: 70% of high-value patents remain concentrated in developed economies (WIPO, 2022), stifling inclusive innovation. Blockchain tokenization now emerges as a *social-technical disruptor*, promising to democratize IP access while reconfiguring innovation ecosystems (Tapscott & Tapscott, 2016; Yermack, 2019). Yet critical gaps persist between theoretical potential and empirical validation, particularly regarding liquidity dynamics, regulatory frameworks, and equitable participation (Gans & Catalini, 2016; Bonnet & Teuteberg, 2023).

The ERC-20 standard (Buterin, 2013; Wood, 2019) enables this transition by transforming static patents into fractionalized, tradable digital assets. By encoding ownership and royalty distribution into self-executing smart contracts (Szabo, 1997; Atzei et al., 2017), blockchain mitigates intermediation costs and enhances transactional transparency (Atzori, 2015; Zhang et al., 2019). Early research highlights tokenization's capacity to unlock liquidity for traditionally illiquid assets (Schär, 2021; Brauneis et al., 2021), yet its application to IP markets remains nascent. Existing studies predominantly explore conceptual frameworks (Candelin-Palmqvist et al., 2012; Kouhizadeh & Sarkis, 2018) or isolated technical implementations (Alqarni, 2024; Yuan et al., 2024), neglecting three key dimensions:

1. *Market dynamics*: How token design and investor behaviour shape liquidity across sectors (Cong et al., 2020; Voigt et al., 2020);
2. *Policy foresight*: Regulatory models enabling inclusive participation while mitigating risks like speculative hoarding (Malinova & Park, 2017; Savelyev, 2017);
3. *Innovation equity*: Mechanisms to bridge the "tokenization divide" between high- and low-liquidity sectors (Swinkels, 2023; Schwarz, 2022).

This study addresses these gaps through large-scale socio-technical forecasting. We simulate 20 million transactions across tokenized patents to: i) Quantify liquidity enhancement in technology, healthcare, and industrial sectors (§3.1–3.3); ii) Model investment fund viability under equal-weight, momentum, and mean-reversion strategies (§2.4, §3.3.3); iii) Forecast liquidity evolution using machine learning (§2.5, §3.5); and iv) Propose governance frameworks for *inclusive innovation* (§4).

Our contributions actively reshape socio-technical futures in three key dimensions: i) Empirically validating tokenization's role in reducing IP inequality (cf. Kim, 2020; Gan et al., 2020); ii) Introducing a simulation methodology for forecasting decentralized market behavior (cf. Malinova, 2016; Ciriello, 2021); iii) Mapping policy pathways to avoid replicating historical centralization in emergent Web3 ecosystems (Schär, 2021; Savelyev, 2017).

By bridging blockchain's technical promise with actionable socio-economic foresight, this work illuminates how tokenization could reconfigure innovation, not merely as an asset class, but as a participatory engine for global problem-solving.

The remainder of the paper is structured as follows: Section 2 presents the data and methodology, Section 3 outlines the results and analysis, Section 4 offers a discussion, and Section 5 concludes the study.

2. Data and Methodology

2.1 Data

The study uses data from Google Patents and the S&P Capital IP Pro Database. Google Patents provides critical information on patents, such as unique patent IDs, titles, assignees, inventors, and key dates such as filing, publication and grant. It also provides links to entire patent filings and descriptions of typical figures. Company data, on the other hand, was obtained from the S&P Capital IP Pro database, which focusses on large firms in the S&P 500 index. Market capitalization, total revenue, industry and sector classification are some of the key financial measures obtained. We collected at least 10 patents for each firm from the top 30 corporations in three primary industries: technology, healthcare and manufacturing industry.

2.1.1 Data Overview and Pre-processing

This study focuses on high-innovation industries to evaluate patent tokenization's potential to improve liquidity and fractional ownership. Given patents' role in IP protection and innovation incentives (Moser, 2013), their tokenization raises critical IP rights considerations. The dataset tracks patents and firms using patent IDs as primary keys.

id	US-2024249379-A1
title	Automatic cropping of video content
assignee	Apple Inc.
inventor/author	May-Li Khoe, Avi E. Cieplinski, David Joshua HART
priority date	3/9/15
filing/creation date	4/5/24
publication date	7/25/24
grant date	NaN
result link	https://patents.google.com/patent/US2024024937...
representative figure link	https://patentimages.storage.googleapis.com/a1...
Exchange: Ticker	NASDAQGS:AAPL
Ticker	AAPL
Trading Currency	USD
Market Capitalization (\$000)	3436895169.0
Total Revenue (Last Twelve Months) (\$000)	385603000
Primary Industry	Technology Hardware, Storage and Peripherals
Country / Region Name	USA
Sector	Information Technology
Name: 0, dtype: object	

Figure 1: Patent metadata for US-2024249379-A1. The table represents for a given patent: title, assignee, inventor/authors and for each one of them the financial metrics of market capitalization and total revenue. (Source: S&P Capital IQ Pro, Google Patents)

The dataset contains patent ID, title, assignee, inventor, different important dates (filing, publication and grant dates) and financial indicators. Furthermore, industry and sector classifications provide context for investigating how patents are dispersed throughout various economic sectors. Preparation of the dataset ensured that the patent data was structured for tokenization, permitting the creation of ERC-20 tokens reflecting fractional ownership of these patents. This change was required to ensure that the data could be compared accurately across firms and industries. Our study is based on the resulting dataset of 298 distinct patents submitted by 30 businesses. The dataset is designed to investigate the viability of patent tokenization and evaluate its liquidity in simulated market conditions. By merging individual patent information with financial performance data, it provides a complete view of the present spectrum of patent tokenization opportunities across several industries.

2.2 Tokenization Process

Tokenization converts patent rights into digital tokens on a blockchain, enabling fractional ownership and enhanced liquidity. This transformation is particularly impactful for intellectual property (IP), as it unlocks new monetization avenues for traditionally illiquid assets. By fractionalizing patent ownership, tokenization broadens investor participation and improves market accessibility. Additionally, it streamlines licensing and royalty distribution while facilitating novel financial instruments (e.g., tokenized patent funds). As outlined earlier, blockchain-based IP management offers significant potential to enhance transparency, security, and efficiency.

2.2.1 Tokenization Methodology

Tokenizing patents into ERC-20 tokens requires several key procedures, all of which contribute to the tokens' overall design and functioning. ERC-20 is the most widely used standard for token issuance on the Ethereum blockchain, enabling interoperability, convenience and robust ecosystem. The ERC-20 standard was chosen because of its proven, tested structure, which ensures that the tokens can be easily integrated into existing decentralized apps (dApps) and marketplaces.

The initial step in the tokenization process is to pick the patents that will be represented as digital tokens. Each patent is assigned a unique identification, which often corresponds to its official patent ID and is critical for tokenization and subsequent trade operations. This unique identification ensures that each token is individual and can be traced back to the underlying intellectual property asset.

Patent ID: US-2024249379-A1, Symbol: AUT, Initial Value: \$31.45, Total Supply: 1592108
Patent ID: US-2024248597-A1, Symbol: POR, Initial Value: \$30.69, Total Supply: 1237061
Patent ID: US-2024240960-A1, Symbol: DEV, Initial Value: \$28.69, Total Supply: 849467
Patent ID: US-2024244097-A1, Symbol: ALA, Initial Value: \$26.49, Total Supply: 1850312
Patent ID: US-2024248515-A1, Symbol: MET, Initial Value: \$31.24, Total Supply: 1981072
Patent ID: US-2024244531-A1, Symbol: MET, Initial Value: \$28.57, Total Supply: 1789837
Patent ID: US-2024225548-A1, Symbol: SEN, Initial Value: \$27.44, Total Supply: 698423
Patent ID: US-2024233752-A1, Symbol: CON, Initial Value: \$28.94, Total Supply: 1136894
Patent ID: US-2024214995-A1, Symbol: MET, Initial Value: \$27.01, Total Supply: 816435
Patent ID: US-2024210997-A1, Symbol: ELE, Initial Value: \$31.15, Total Supply: 687164
Patent ID: US-2024244252-A1, Symbol: ENC, Initial Value: \$30.51, Total Supply: 616840
Patent ID: US-2024195973-A1, Symbol: ADA, Initial Value: \$29.49, Total Supply: 1828547
Patent ID: US-2024160795-A1, Symbol: PER, Initial Value: \$29.31, Total Supply: 1055677
Patent ID: US-2024155147-A1, Symbol: CON, Initial Value: \$30.44, Total Supply: 1432448
Patent ID: US-2024135098-A1, Symbol: INT, Initial Value: \$28.16, Total Supply: 1571753
Patent ID: US-2024080475-A1, Symbol: SIG, Initial Value: \$30.56, Total Supply: 1027602
Patent ID: US-2024080485-A1, Symbol: RED, Initial Value: \$29.27, Total Supply: 1497312
Patent ID: US-2024061543-A1, Symbol: APP, Initial Value: \$27.04, Total Supply: 1061131

Figure 2: This table displays several patents identifiable by their unique IDs, connecting symbols, initial values in USD, and total token supply for each patent.

Once a patent is identified, its name and title will be used to generate the token name and symbol that will allow the patent to be recognized in the blockchain ecosystem. The token sign is created from the initial characters of the patent's name, allowing users to recognize it in transactions and listings.

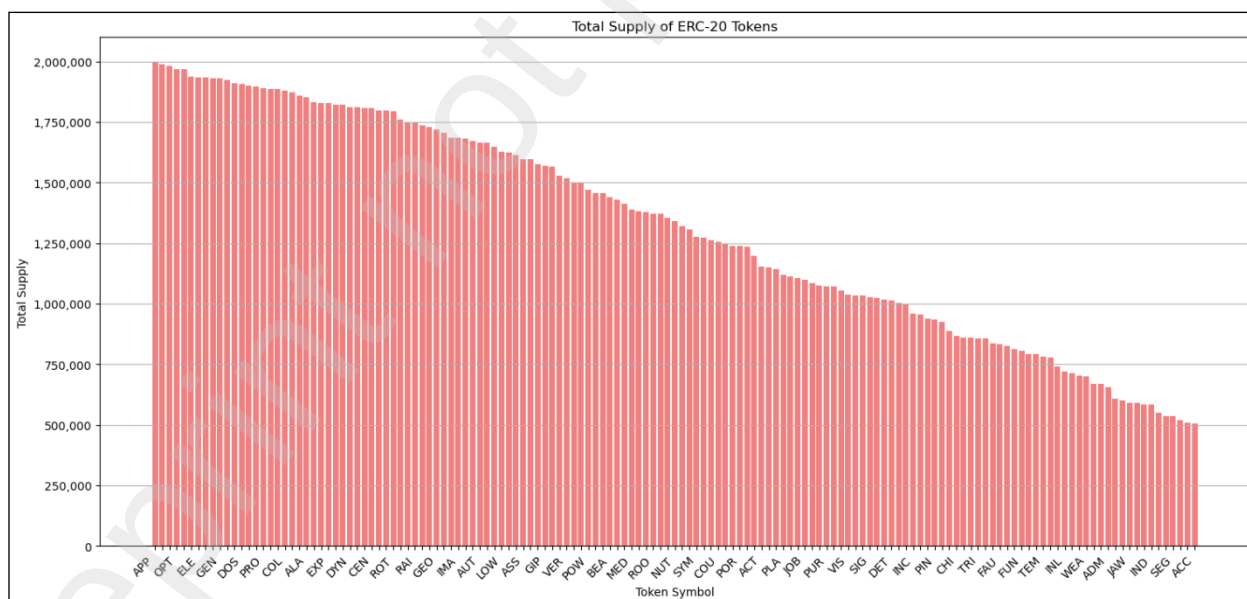


Figure 3: The Bar graph of overall supplies corresponding to different ERC-20 tokens, represented by their respective symbols.

The process also includes calculating the total supply of each ERC-20 token, which is the maximum number of tokens that can be manufactured and distributed for each patent. In this simulation, the entire supply is randomly distributed within a range of 500,000 to 2,000,000 tokens. This randomization seeks to imitate real-world unpredictability, in which an asset's supply has a substantial impact on market dynamics. A higher supply may signify greater availability, potentially resulting in cheaper pricing, whereas a smaller supply may produce scarcity, increasing demand and value.

2.2.2 Token Valuation and Supply Mechanism

The initial value of each ERC-20 token is derived from the patent's market capitalization, converted from thousands to exact dollar values. We apply a natural logarithm transformation to normalize values across patents and compress the price range. This logarithmic market cap is then adjusted by a random factor (0.9-1.1) to simulate market volatility. Token supply is randomized within a predetermined range to create a diverse ecosystem, enabling examination of how supply-demand dynamics affect value. This approach facilitates the exploration of various market scenarios while maintaining realistic valuation parameters. As previously stated, the overall supply of each token is randomized within a predetermined range. The goal of this choice is to establish a dynamic and diverse token ecosystem in which various tokens experience variable levels of scarcity and demand. By limiting the overall supply to a specific range, the tokenization process enables the exploration and study of various market situations based on the relationship between supply, demand, and value.

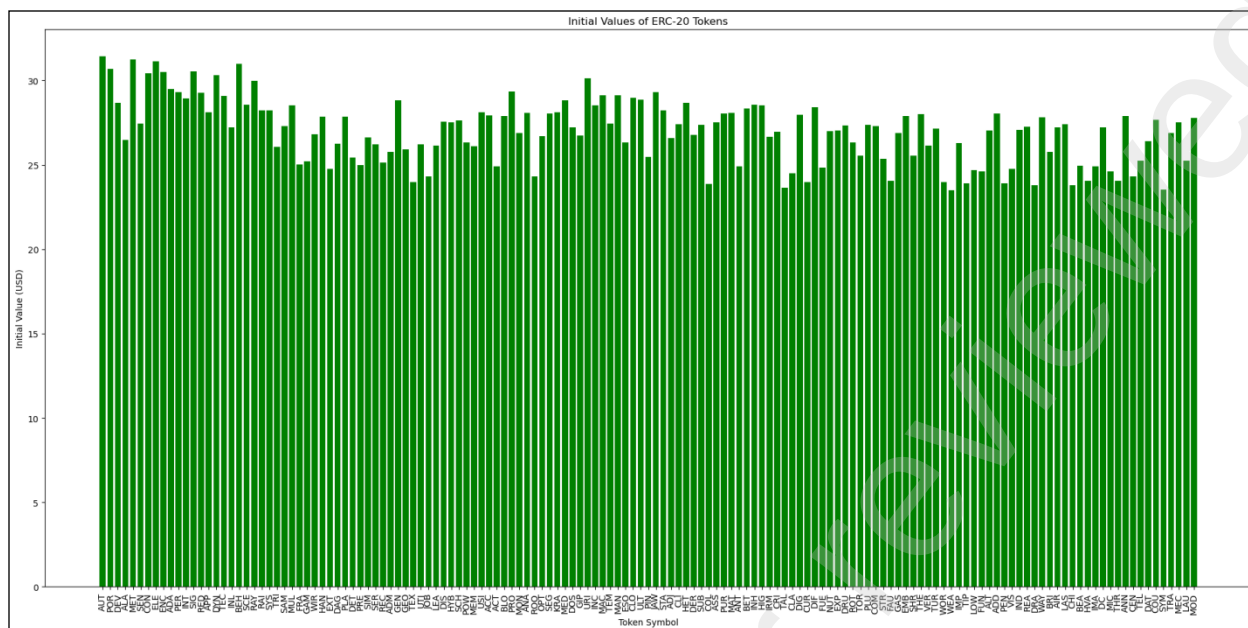


Figure 4: The bar chart shows the initial market value in USD for numerous ERC-20 tokens, as well as the change in value of the tokens at the time of their entry into the market.

2.2.3 Random Factors and Assumptions

Incorporating random factors and assumptions into the tokenization process is critical for producing a realistic and adaptive simulation of how digital assets would behave in a real-world market. The random factor applied to market capitalization before determining the initial token value accounts for the inherent unpredictability in asset valuation, recognizing that market perceptions, investor sentiment, and external economic conditions can all influence asset value, thereby introducing variability that mimics real-world dynamics.

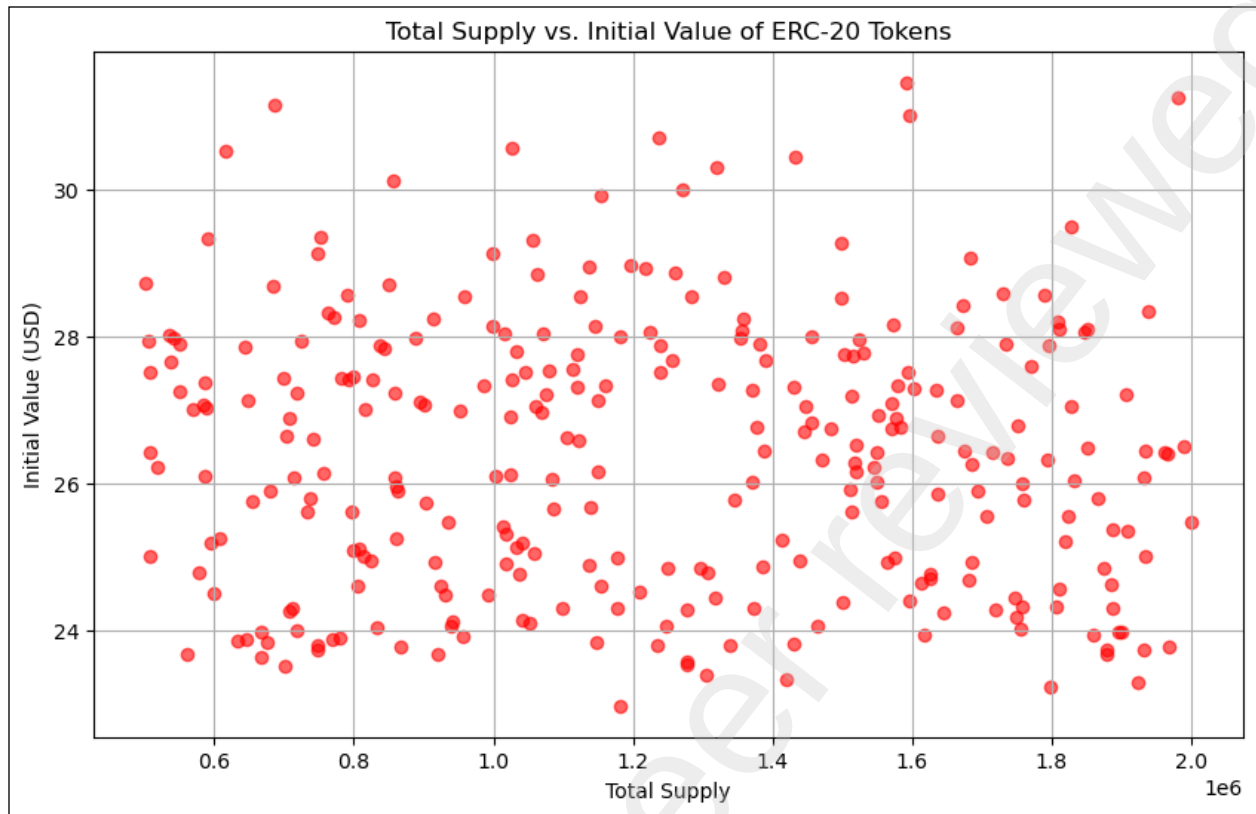


Figure 5: The scatter plot depicts the entire supply versus the market value of the ERC-20 token, which aids in detecting any connection between a token's initial value and supply into the market.

Furthermore, the randomization of total supply reflects the different degrees of interest and demand for each patent. A patent with a higher perceived value may have a bigger token supply, indicating greater availability and possible liquidity, whereas a specialized patent may have a lower quantity, implying exclusivity and higher value per token. These assumptions are critical for replicating a diversified market environment in which different tokens can display distinct behaviors based on their assigned qualities.

The random factors and assumptions utilized in this simulation were carefully designed to mimic realistic market conditions. They allow for a full examination of how tokenized intellectual property may behave in a decentralized market, providing significant insights into the possible benefits and limitations of such a system. This simulation assesses the viability of using ERC-20 tokens to boost liquidity and allows for fractional ownership of patents, democratizing access to intellectual property assets.

2.3 Market Simulation

In any financial market, investor behavior has a significant impact on market dynamics. To simulate a realistic market for tokenized patents, we created several investor profiles that represent common trading techniques. These profiles were created to imitate a variety of real-market habits, such as long-term holding, frequent trading, and market-making.

2.3.1 Investor Profiles

We created three separate investor profiles, each reflecting a particular trading technique and behavior, to capture a wide range of market characteristics.

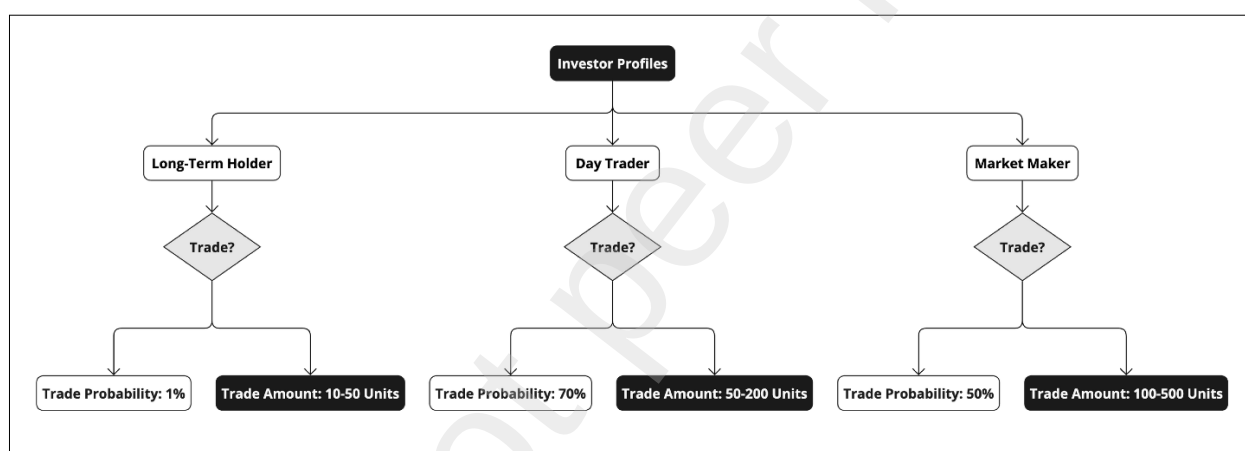


Figure 6: The flowchart displays several investor profiles, such as Long-Term Holders, Day Traders, and Market Makers, as well as their trading habits, to demonstrate how strategies affect trade volume and market participation.

We model three distinct trader types with characteristic behaviours: i) *Long-term holders* exhibit low-frequency trading (1% probability per period) with conservative transaction sizes (10-50 tokens), prioritizing capital appreciation over active trading; ii) *Day traders* demonstrate high activity (70% trading probability) with larger transaction volumes (50-200 tokens), capitalizing on short-term price fluctuations and iii) *Market makers* maintain market liquidity through balanced buy/sell orders (50% trading probability) with the largest transaction sizes (100-500 tokens), ensuring continuous market depth.

This typology captures the essential trading dynamics while maintaining market stability through complementary behaviours.

2.3.2. Experimental Setup

The simulation initialized 50 investor addresses, with each randomly assigned one of three trading profiles (long-term holders, day traders, or market makers) to simulate natural market participant diversity. This randomization protocol ensured heterogeneous trading behaviors representative of real market conditions.

I. Token Distribution

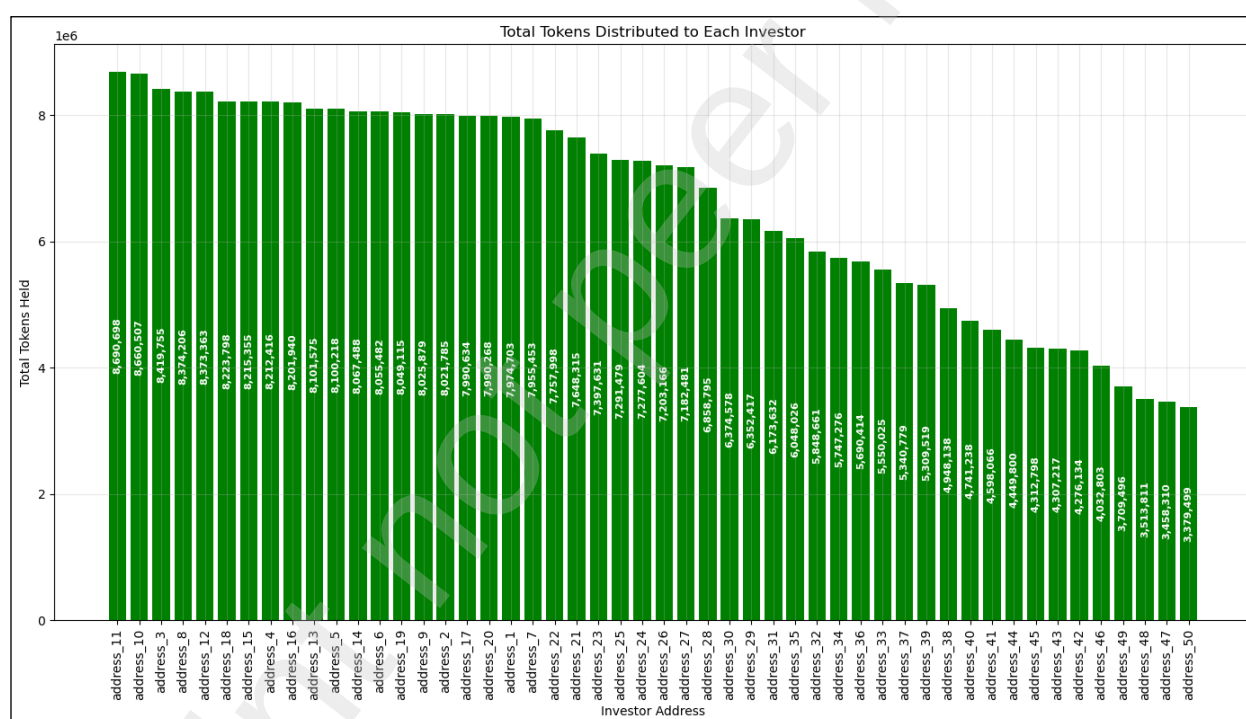


Figure 7: The bar chart depicts the distribution of total tokens held by various investors, displaying their unique addresses and the concentration of token ownership, so providing insight into individual investors' influence in the token ecosystem.

Before the market simulation could begin, the initial stock of tokens needed to be allocated to investors. The distribution procedure was random, with each investment receiving a portion of the entire token supply. This technique ensured that the market began with diverse ownership

structures, like the initial distribution period in a real market where different investors acquire tokens based on their investing plans.

II. Transaction Simulation

The transaction simulation is built to manage up to 20 million transactions, therefore producing a rich dataset for study to replicate the complexity and dynamics of a real-world market for tokenized patents. Every iteration in the simulation reflects either a possible trade or a choice not to trade depending on the behavior of randomly chosen investors from a varied pool. From long-term holders to day traders, each with different trading probability and transaction amounts that represent their market strategy, these investors span.

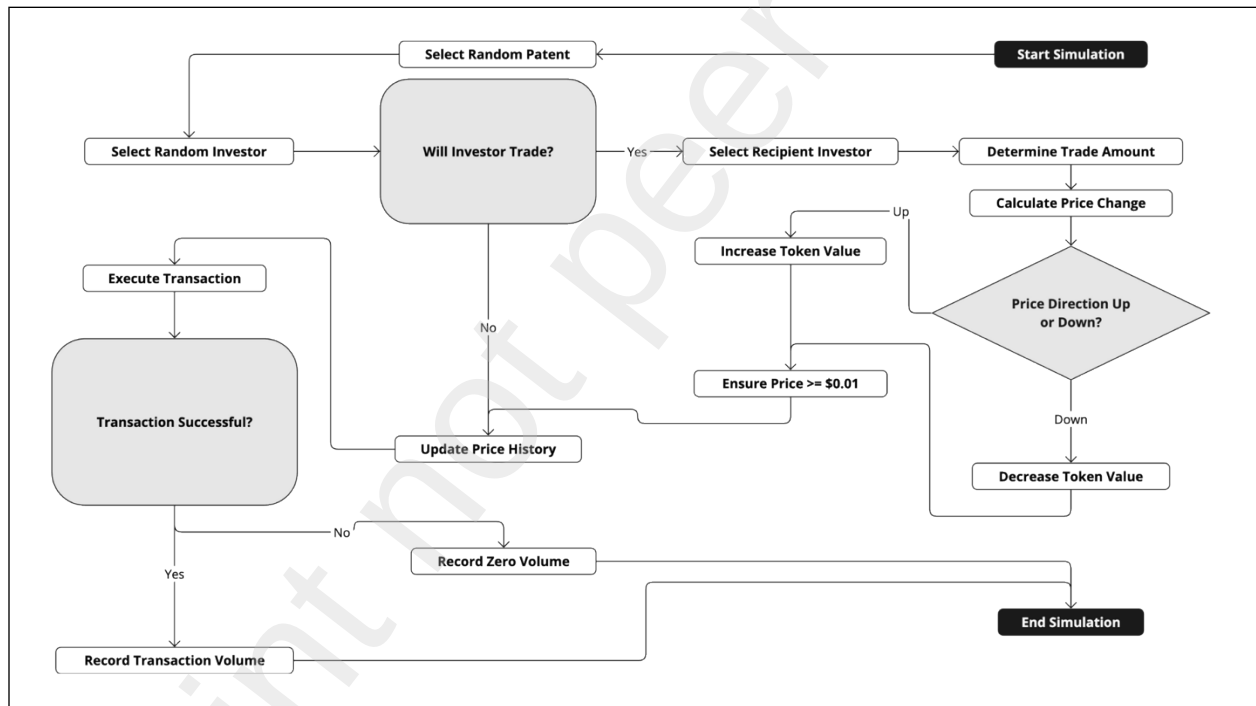


Figure 8: The flowchart illustrates how the ERC-20 token trading simulation works. It begins with investor and patent selection at random, then moves on to trading decisions, trade amount, and price adjustments. In addition, it provides price history updates based on transaction success and transaction volumes.

Each transaction randomly selects a patent token to simulate shifting market attention. Token prices adjust based on i) Transaction size relative to total supply, and ii) Randomly determined direction ($\pm$) to capture market volatility.

The simulation records all transaction volumes and corresponding price changes, generating a comprehensive dataset for analysis. This enables evaluation of key market metrics, including liquidity, price impact of trades and overall stability, providing insights into the dynamics of tokenized patent markets under varying trading conditions.

III. Visualization of Market Dynamics

The simulation results are visualized to clearly show the evolution of market dynamics over time, providing important insights into price changes and trading activity. Each token's historical pricing is plotted, recording value changes as they occur during the simulation. The figure below illustrates the volatility of each token and how different investment behaviors affect price stability. Transaction volumes serve as a key indicator of trading intensity per token, enabling assessment of both market liquidity and relative demand across tokenized patents. These metrics, combined with price data, provide a comprehensive view of market dynamics, explaining the mechanisms of price discovery and liquidity formation.

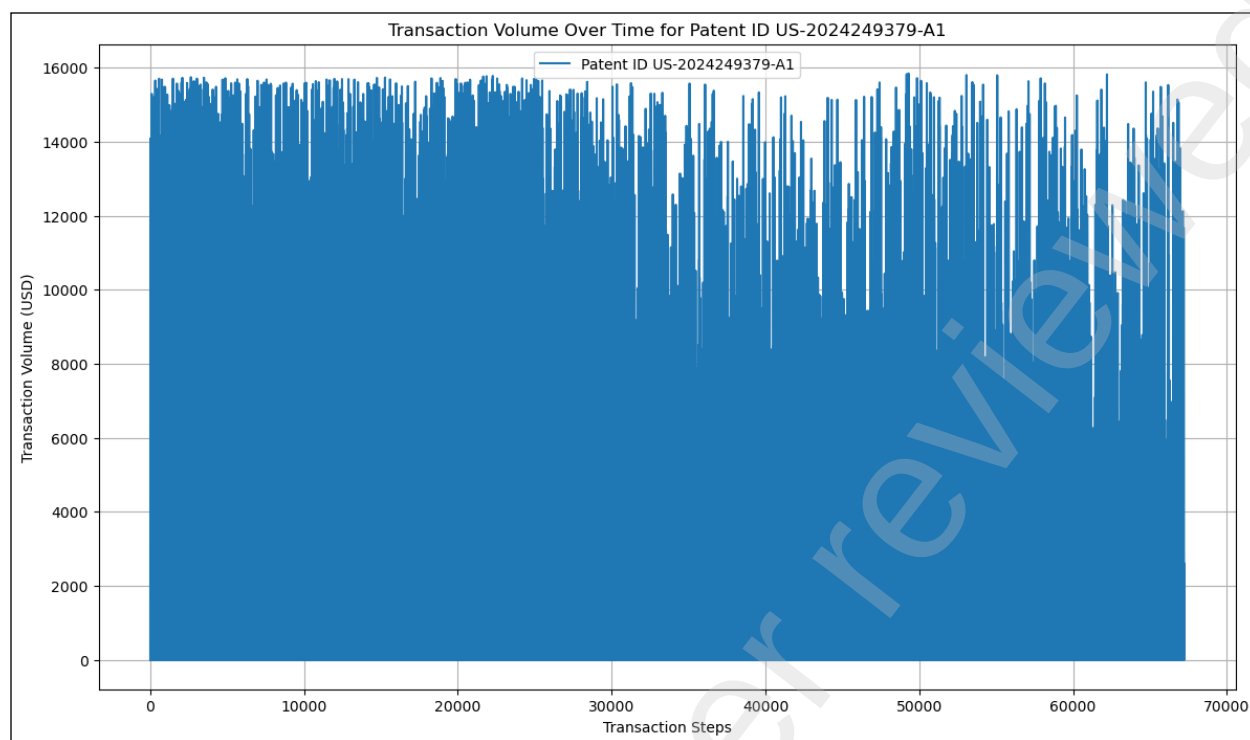


Figure 9: The figure illustrates the USD transaction volume for US-2024249379-A1, showcasing variations in trading activity and investor interest over a period, which is essential for comprehending the liquidity and market involvement of this tokenized patent.

2.3.3 Liquidity Metrics

Liquidity, a key indicator of market efficiency, reflects the ease of trading assets without significant price changes. In our simulation, we assessed liquidity using price impact, the absolute price change before and after a trade divided by trade size. A low-price impact indicates high liquidity, where large transactions minimally affect prices, suggesting a stable and efficient market.

2.4 Simulation of Investment Fund

This study simulates an investment fund to assess how alternative trading strategies can enhance returns for tokenized intellectual property rights (IPR). We modeled different approaches to evaluate their impact on fund performance under varying market conditions. This section details the token selection process, trading strategies employed, and overall fund performance.

2.4.1 Selection of Tokens for the Fund

The fund's portfolio was initialized by randomly selecting 20 tokens from the simulated market pool. This randomization ensures diversification, mitigating single-asset concentration risk and approximating real-world fund management practices.

Initialized IPR Fund with \$1000000.00 in capital and the following tokens:	
- NUT (NUT):	\$26.45 per token
- AUT (AUT):	\$26.87 per token
- SYS (SYS):	\$27.26 per token
- NUT (NUT):	\$23.70 per token
- GEN (GEN):	\$27.05 per token
- TEC (TEC):	\$26.12 per token
- APP (APP):	\$27.31 per token
- MUL (MUL):	\$25.11 per token
- SCE (SCE):	\$28.32 per token
- PRO (PRO):	\$30.21 per token
- SYS (SYS):	\$27.63 per token
- DYN (DYN):	\$28.59 per token
- ACT (ACT):	\$24.45 per token
- PRO (PRO):	\$26.90 per token
- ADM (ADM):	\$26.89 per token
- UTI (UTI):	\$24.42 per token
- GEN (GEN):	\$25.58 per token
- ESO (ESO):	\$26.41 per token
- CRI (CRI):	\$26.07 per token
- SIG (SIG):	\$29.78 per token

Figure 10: The picture depicts the initial establishment of an IPR fund with a capital of \$1,000,000 distributed across tokens linked with various patents, establishing a baseline for the fund's value allocation among selected tokens.

The fund was initialized with an initial capital of \$1,000,000, held entirely in cash with no token positions. This setup mirrors real-world fund management, where capital is gradually deployed into assets according to the investment strategy. Starting with a cash-only position allowed for controlled and systematic portfolio construction.

2.4.2. Trading Strategies

Three distinct trading strategies were implemented to evaluate different portfolio management approaches: i) *Equal Weight (EW) Strategy*: Allocates capital uniformly across all selected tokens, ensuring balanced exposure and mitigating single-asset risk; ii) *Momentum-Based (MB) Strategy*: Dynamically increases positions in tokens with positive price momentum while divesting underperformers, capitalizing on short-term trend persistence and iii) *Mean Reversion (MR) Strategy*: Takes contrarian positions by acquiring underperforming tokens (anticipating recovery)

and selling overperforming ones (expecting correction), based on long-term price mean reversion tendencies.

Each strategy reflects differing risk-return profiles, from the conservative EW approach to the more aggressive MB and MR frameworks.

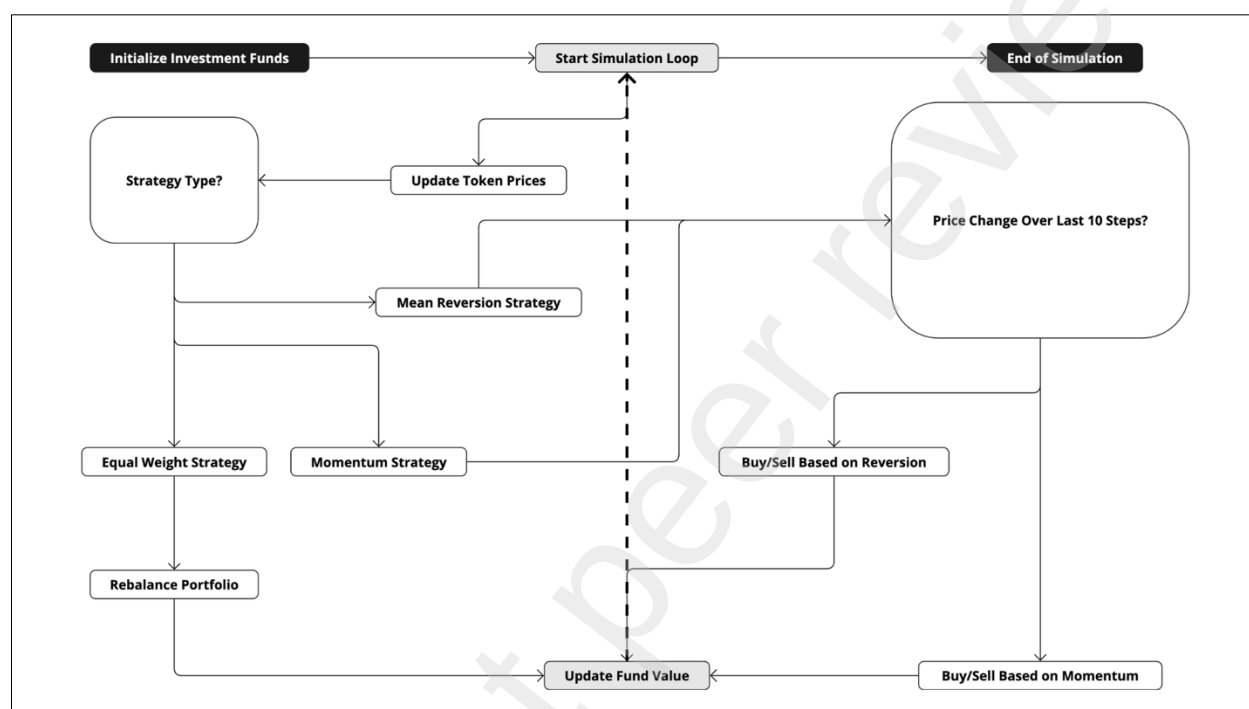


Figure 11: The simulation process for managing investment funds includes initializing funds, choosing a strategy (such as Equal Weight, Momentum, or Mean Reversion), rebalancing the portfolio, updating token prices, and making buy/sell decisions. The flowchart provides a clear overview of how various strategies affect fund value over time, which guides decision-making during the simulation.

Following implementation, we tracked the fund's value over time to evaluate the effectiveness of each strategy. The Figure below demonstrates their comparative performance, revealing distinct growth patterns and risk-return profiles across the three approaches.



Figure 12: The charts present a comparison of the total return and standard deviation for the Equal Weight, Momentum, and Mean Reversion strategies.

2.5 Machine Learning Models

We use machine learning algorithms for predicting the liquidity of tokenized patents, or IPRs, using attributes extracted from simulated market data. Below we discuss the steps involved, i.e. feature engineering, model selection with cross validation and grid search, model fitting and validation and feature importance analysis.

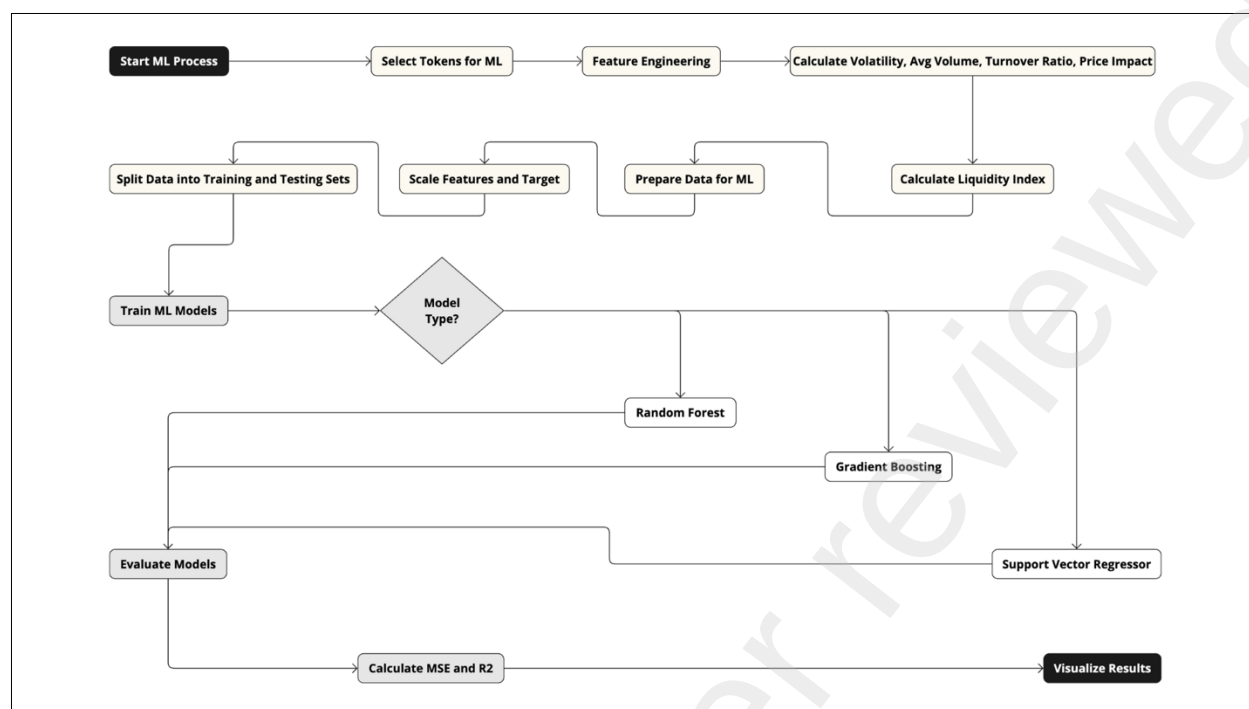


Figure 13: Token liquidity index prediction by machine learning entails feature engineering, model selection, training, and assessment with Random Forest, Gradient Boosting, and Support Vector Regressor, utilizing MSE and R^2 .

2.5.1 Feature Engineering

Effective feature engineering is critical for model performance, as the predictive power of machine learning algorithms depends heavily on feature quality and relevance. We derived the following liquidity-related features from simulated token price and transaction data:

1. *Volatility (σ)*: Standard deviation of token prices, quantifying market stability. High volatility often correlates with reduced liquidity.
2. *Average Volume*: Mean trading volume per period, where higher values typically indicate greater market depth.
3. *Turnover Ratio*: Total traded volume normalized by token supply, reflecting exchange frequency and liquidity potential.

4. *Average Price Impact*: Measures marginal price change per unit trade, with lower values indicating higher liquidity. It is defined by: $Price\ Impact = \frac{|P_t - P_{t-1}|}{Trade\ Volume}$
5. *Liquidity Index* (Target Variable): A composite metric where higher values denote better liquidity. It is defined by: $Liquidity\ Index = \frac{Turnover\ Ratio}{Average\ Price\ Impact}$

These features capture key market dynamics, ensuring the model learns meaningful relationships for liquidity prediction.

2.5.2 Model Architecture

For comparative analysis, we evaluated three distinct machine learning approaches using default hyperparameters to establish baseline performance:

1. *Random Forest Regressor*: Selected for its dual advantages in handling high-dimensional data while maintaining interpretability through native feature importance quantification. The ensemble approach effectively captures non-linear relationships while mitigating overfitting.
2. *Gradient Boosted Regression (XGBoost)*: Implemented for its superior performance in modelling complex, non-linear dependencies through sequential error correction. The boosting mechanism proves particularly effective when feature-target relationships exhibit hierarchical patterns.
3. *Support Vector Regressor (Linear Kernel)*: Serves as our linear baseline, testing the fundamental hypothesis that liquidity relationships can be captured through a linear mapping. The choice of linear kernel provides interpretable coefficients while establishing a performance floor for more sophisticated models.

This tripartite modelling framework enables comprehensive evaluation across the spectrum of model complexity, from simple linear correlations to advance non-linear representations. The

inclusion of both tree-based and kernel-based methods ensures robust assessment of liquidity prediction approaches.

2.5.3 Model Training and Validation

Data Preprocessing: Prior to model training, all features and the target variable (liquidity index) were standardized using StandardScaler to ensure zero mean and unit variance. This normalization prevents feature magnitude disparities from biasing model learning. The dataset was then partitioned into training (70%) and testing (30%) sets, maintaining a balance between sufficient training data and robust out-of-sample evaluation.

Model Training and Evaluation: Each model was trained on the scaled training data, optimizing parameters to minimize prediction error. Performance was assessed using two key metrics: i) Mean Squared Error (MSE): MSE gauges the average squared difference between the actual and expected values. Lower MSE values indicate better predictive accuracy and ii) Coefficient of Determination (R^2): R^2 is the fraction of the dependent variable variance that could be forecast from the independent variables. Values closer to 1 denote stronger explanatory power. Model outputs were stored for comparative analysis, with the best-performing model selected based on these evaluation metrics.

3. Results and Analysis

3.1 Overview of Token Distribution

Our tokenization strategy prioritizes technology, healthcare and manufacturing, sectors with robust intellectual property frameworks, high innovation activity and scalable disruption potential. This focus aligns with broader tokenization trends, where digitally native and high-impact assets drive blockchain-based fractional ownership adoption.

	Token ID	Title	assignee	Inventor/Author	Exchange: Ticker	Ticker	Trading Currency	Primary Industry	Country / Region Name	Sector	Symbol	Initial Value (USD)	Final Value (USD)	Return (%)
75	US-2023401159-A1	Memory pools in a memory model for a unified c...	Advanced Micro Devices, Inc., Ati Technologies...	Anthony Asaro, Kevin Normoyle, Mark Hummel	NASDAQGS:AMD	AMD	USD	Semiconductors	USA	Information Technology	MEM	23.794025	26.010978	9.317265
181	AU-2024201705-A1	Drug delivery device with sterile fluid flowpa...	Amgen Inc.	Yasaman DAMESTANI, Lawrence S. Ring	NASDAQGS:AMGN	AMGN	USD	Biotechnology	USA	Health Care	DRU	25.610630	27.485014	7.318775
48	US-2024168949-A1	Predicting future quiet periods for materializ...	Oracle International Corporation	Murali Thiagarajan, Praveen T.J. Kumar	NYSE:ORCL	ORCL	USD	Systems Software	USA	Information Technology	PRE	25.003688	26.513859	6.039793
192	US-2024232222-A1	Systems and methods for synchronizing nodes of...	Intuitive Surgical Operations, Inc.	Brian D. Hoffman, Joshua Radel	NASDAQGS:ISRG	ISRG	USD	Health Care Equipment	USA	Health Care	SYS	27.981490	29.625382	5.874930
78	US-2023351187-A1	Accelerating neural networks with one shot ski...	Advanced Micro Devices, Inc.	Arun Coimbatore Ramachandran, Chandra Kumar RA...	NASDAQGS:AMD	AMD	USD	Semiconductors	USA	Information Technology	ACC	27.941341	29.552938	5.767787
213	AU-2022268366-A1	System and method for worksite management	Caterpillar Inc.	Praveen Halepatali, Vamsi Kalyan Putta, Vijay ...	NYSE:CAT	CAT	USD	Machinery and Equipment: Construction and Heav...	USA	Industrials	SYS	28.232728	29.814220	5.601627
51	US-2024256409-A1	Server clustering in a computing-on-demand sys...	Salesforce, Inc.	Emily Katherine Witt, Richard Cooke, William T...	NYSE:CRM	CRM	USD	Application Software	USA	Information Technology	SER	26.230685	27.672684	5.497374
127	US-2024206874-A1	Adjustment to transfer power parameters to imp...	Cilag GmbH International	Frederick E. Shelton, IV, Patrick L. Creamer	NYSE:JNJ	JNJ	USD	Pharmaceuticals	USA	Health Care	ADJ	26.606465	28.052870	5.436291
223	US-11773786-B2	Fundamental gear system architecture	Rtx Corporation	William G. Sheridan	NYSE:RTX	RTX	USD	Aerospace and Defense	USA	Industrials	FUN	24.598256	25.927870	5.405317
230	US-2024257287-A1	Computing system configuring destination accel...	Uber Technologies, Inc.	Yuhki Yamashita, Didier Patrick Hilhorst, Brya...	NYSE:UBER	UBER	USD	Application Software	USA	Industrials	COM	26.127084	27.474234	5.156143

Figure 14: The table contains an in-depth summary of tokens associated with various patents, including their initial and ultimate values, return percentages, and sector information, as well as their ID, title, assignee, and financial data.

We show that technology sector patents, particularly those held by firms such as AMD and Oracle, exhibit higher initial token valuations, underscoring the strategic and economic significance of intellectual property in this sector. Healthcare and Industrials-related patent tokens also command substantial, albeit comparatively lower, starting values. This suggests that while innovation potential contributes to valuation, sector-specific dynamics and corporate market positioning play a pivotal role in price determination.

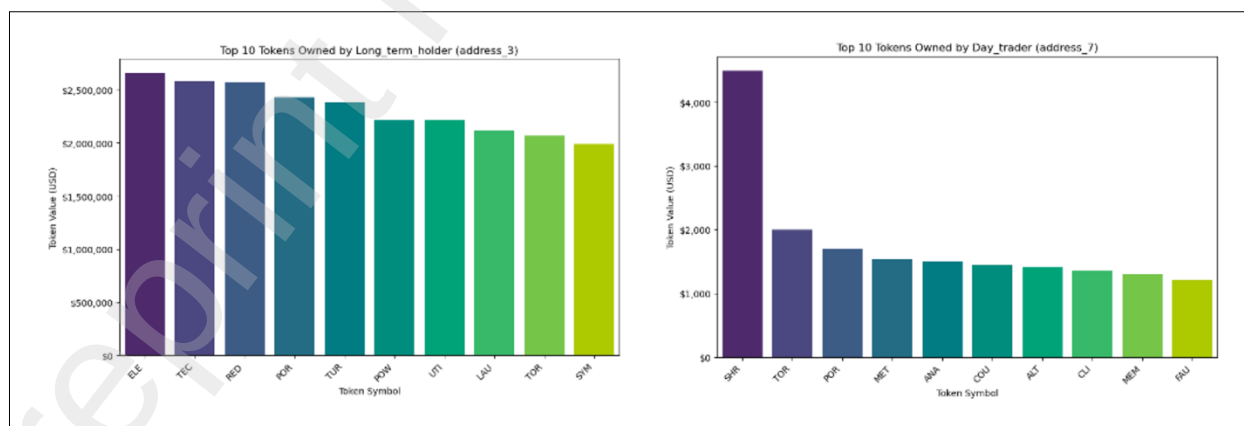


Figure 15: The chart displays the top ten tokens owned by various investor profiles, emphasizing their investment methods and token preferences. Long-term holders prefer tokens with high stability and

long-term growth potential, whilst day traders prioritize volatility and short-term trading opportunities. These charts represent risk tolerance and market behavior.

The figure above reveals distinct patterns in token holdings across investor profiles, shaped by their time horizons and trading strategies. Long-term investors demonstrate a preference for stability, concentrating their portfolios in tokens such as ELE, TEC, and RED, which are predominantly associated with established technology and industrial firms. In contrast, day traders exhibit a more diversified allocation, balancing liquidity needs with upside potential. For instance, their exposure to SHR suggests a focus on high-visibility assets with short-term return potential, likely capitalizing on perceived market inefficiencies and arbitrage opportunities. Meanwhile, their diversified holdings in TEC, DET, and COL reflect a broader tactical approach to exploiting sector-specific opportunities.

3.2. Market Simulation Results

The market simulation reveals distinct price behaviors among the selected tokens over the observed period, as illustrated in the line graph. Notably, tokens such as US-2024249379-A1 and US-2024248597-A1 demonstrated consistent price fluctuations, characterized by low volatility levels of 1.58% and 2.05%, respectively, along with narrow standard deviations of \$ 0.13 and \$0.21. This stability suggests that these tokens were perceived as reliable investment options, likely due to sustained market demand and investor confidence in their underlying value.

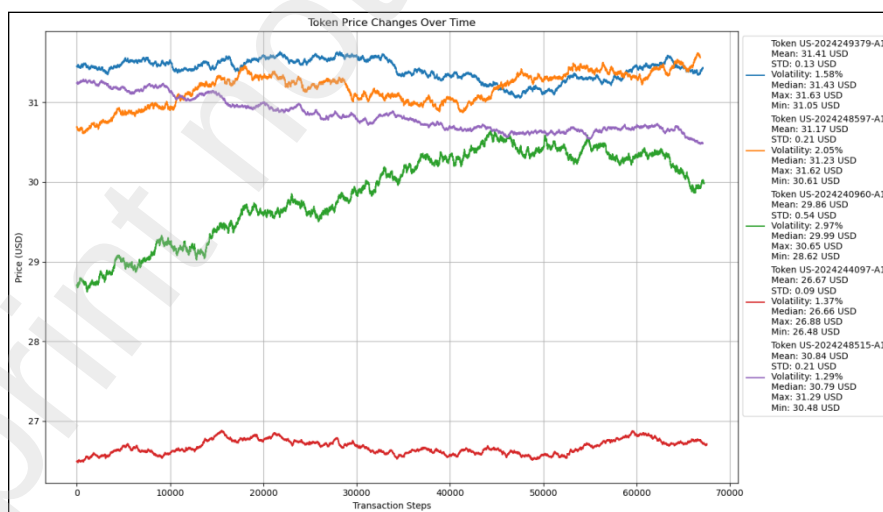


Figure 16: The figure illustrates the price fluctuations of tokens over time, with each token's trend denoted by a unique colour. It highlights discrepancies in price trajectory and overall performance during the observed period, providing insights into market behavior, stability, and growth levels.

Tokens with higher volatility rates of 2.97% and 1.37%, respectively, exhibited more pronounced price fluctuations compared to their lower-volatility counterparts. Specifically, US-20242409-A1 and US-2024244097-A1 demonstrated greater price instability, as reflected in their wider standard deviations of \$0.54 and \$0.09, respectively. These broader deviations suggest erratic market behavior, potentially driven by unforeseen liquidity events, shifts in market sentiment, or speculative trading activity. While such volatility introduces heightened investment risk, it may also present greater short-term profit opportunities, making these tokens particularly attractive to traders pursuing rapid returns despite their inherent unpredictability.

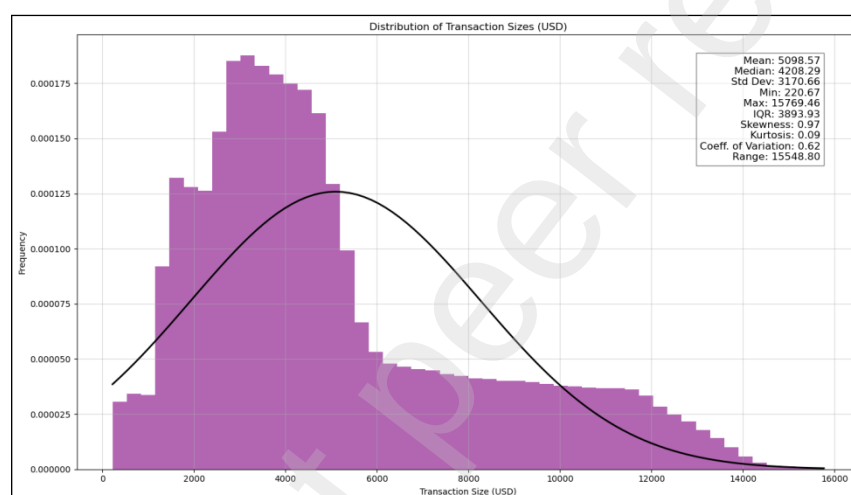


Figure 17: The histogram indicates a right-skewed distribution of transaction sizes throughout 298 tokens, with the majority lying in the lower range. Key statistical variables such as mean, median, standard deviation, and skewness provide an entire overview of the distribution.

These observations can be further validated using the transaction volume histogram, which reveals the distribution of trade sizes. The histogram demonstrates that most transactions occur at moderate volumes, suggesting a market dominated by consistent, albeit relatively small-scale, trading activity. The declining frequency at higher volume-tiers indicates either diminishing liquidity for larger trades or a deliberate reduction in large-volume transactions as market participants seek to minimize price impact. This pattern may reflect strategic behavior by investors to avoid substantial market disruption when executing sizable orders.

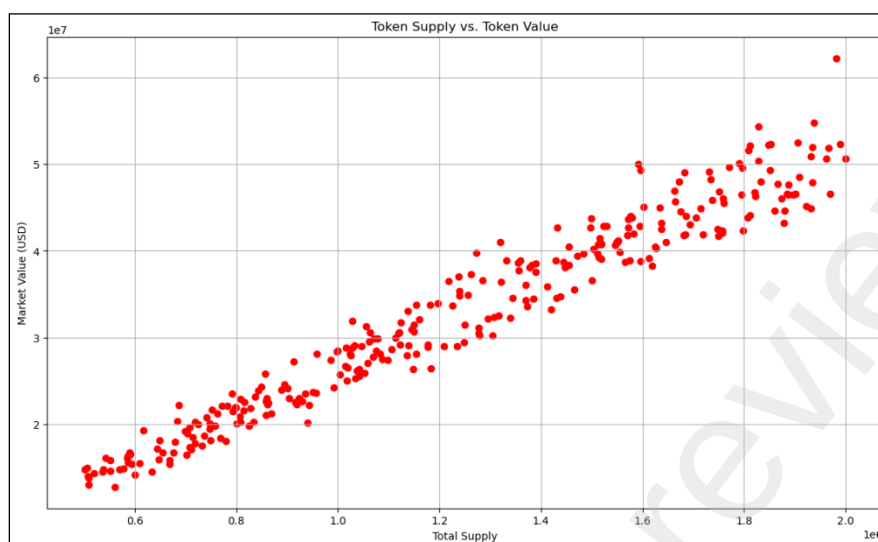


Figure 18: The scatter plot indicates an upward correlation between the total supply of tokens and their related market value for 298 tokens.

The scatter plot reveals a strong positive correlation between token supply and market value, suggesting that increased supply—often indicative of wider distribution—may enhance liquidity and market confidence, thereby boosting valuation. However, outliers demonstrate that additional factors, including investor sentiment and strategic market behavior, significantly influence market value independent of supply.

3.3 Liquidity Analysis

Liquidity is a key determinant of asset viability, reflecting how easily an asset can be traded without major price changes. In our study, we analyze multiple liquidity metrics, including turnover ratio, market depth, bid-ask spread and price impact per trade size, to assess the tradability of patent-linked tokens. These metrics inform investors about trading ease and costs, shaping their decisions.

3.3.1 Liquidity Metric Outcomes

I. Turnover Ratio Analysis: The turnover ratio, the proportion of a token's total supply traded within a given period, serves as a key liquidity indicator. A higher ratio suggests more frequent trading, enhancing liquidity. Our analysis, based on transaction volumes relative to total supply,

reveals turnover ratios between 20% and 47%, indicating moderate to high liquidity. Notably, some tokens exceeded 40%, reflecting strong liquidity and lower trading friction. These findings suggest that highly traded tokens are more attractive to investors seeking minimal price impact when entering or exiting positions.

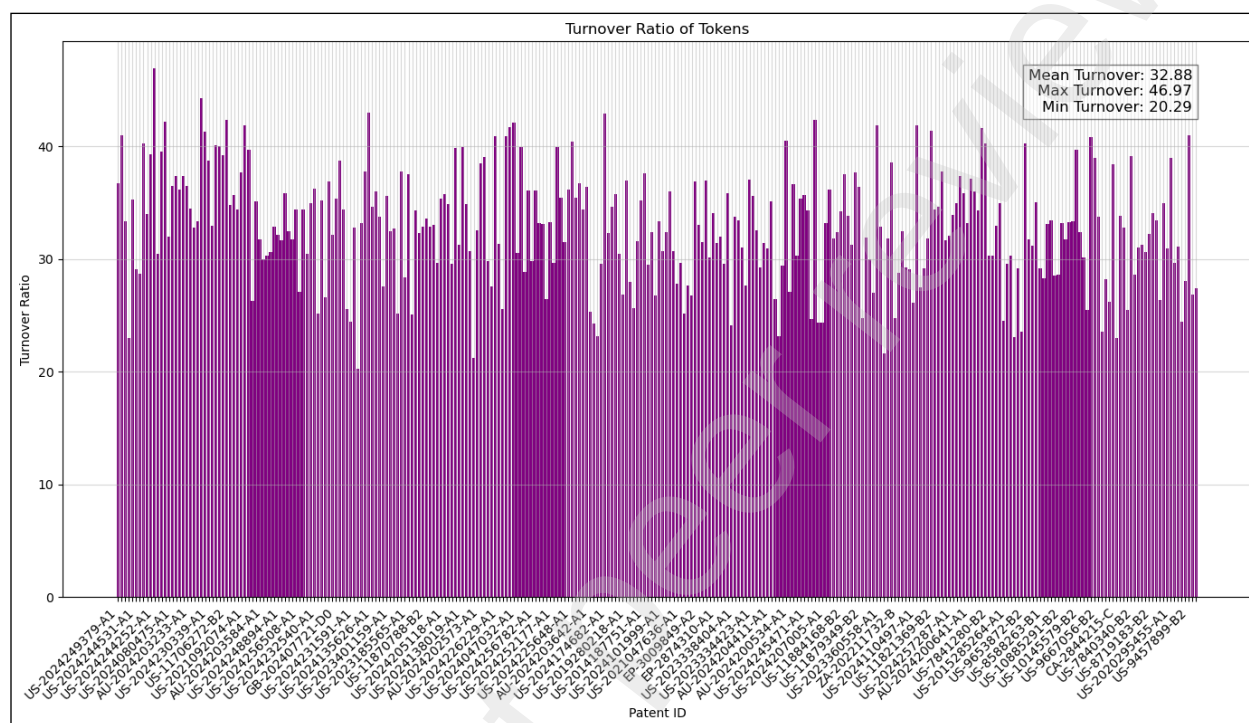


Figure 19: The chart shows the turnover ratios of tokens related to various patents, which indicate their trading activity levels.

II. Market Depth Analysis: Market depth, measured by the volume of buy and sell orders at different price levels, indicates a market's ability to absorb large trades without substantial price fluctuations. Our analysis reveals market depths ranging between 85,000 and 165,000 tokens, with some tokens exhibiting even greater depth. This suggests strong liquidity, enabling large transactions with minimal slippage, a critical feature for investors seeking low-cost execution. By aggregating order book data across price levels, we assess each token's market resilience, finding that deeper markets provide a more stable trading environment, appealing to both retail and institutional investors.

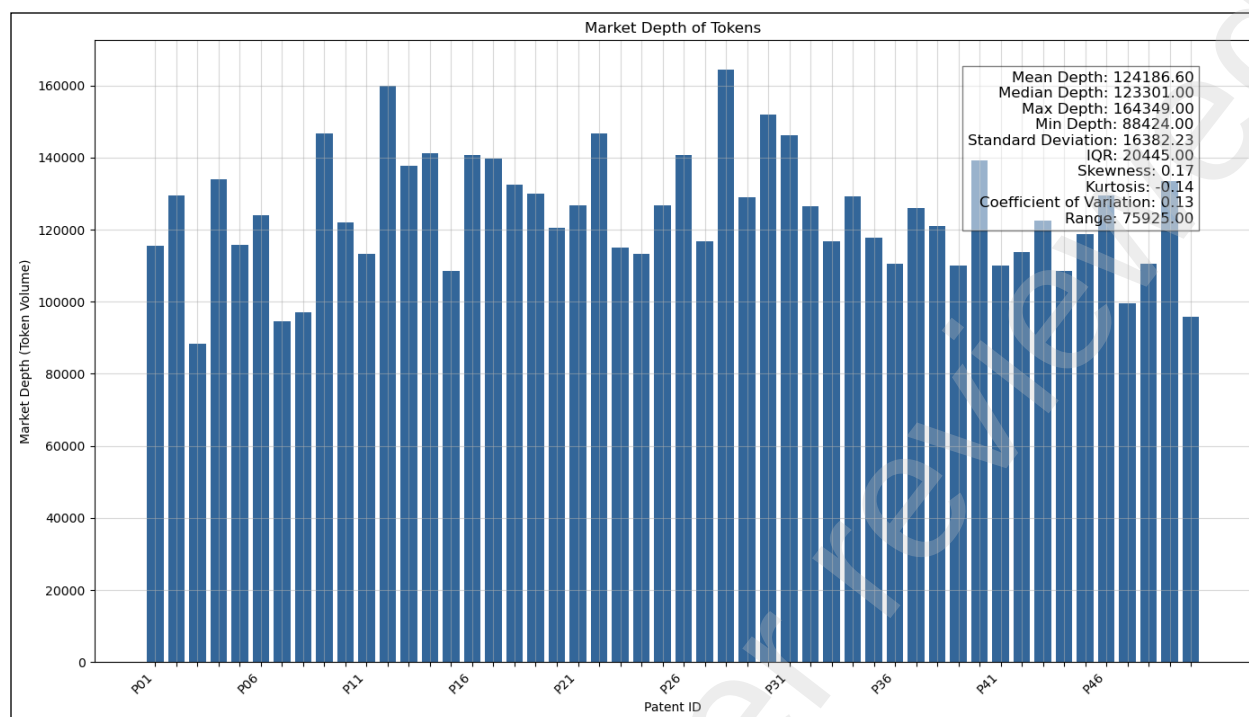


Figure 20: The graph depicts the market depth of tokens across various patents, offering crucial information about their liquidity and trading demand.

III. Bid-Ask Spread Analysis: The bid-ask spread, representing the difference between buying and selling prices, serves as a direct measure of trading costs and inversely reflects liquidity. Our analysis, modelled after over-the-counter market-making dynamics, reveals spreads ranging from 2.0 to 17.0 across tokens. First, tighter spreads (~\$2.0) indicate higher liquidity, making these tokens more attractive for frequent trading due to lower transaction costs. Second, wider spreads suggest reduced liquidity, as higher trading costs deter market participation.

This metric is particularly crucial for active traders and investors prioritizing rapid position entry and exit with minimal cost.

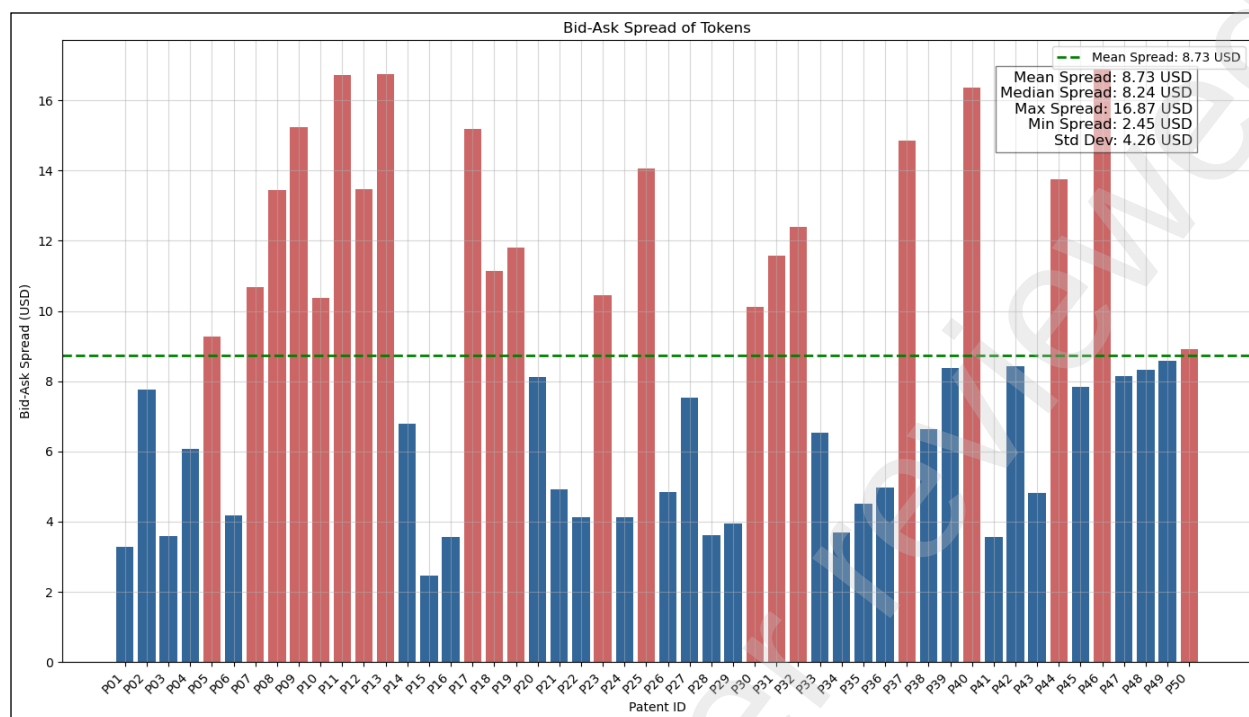


Figure 21: The chart shows the spread of bid-asked tokens across different patents, suggesting liquidity and market efficiency, with an average spread of 8.73 USD.

IV. Price Impact Analysis: Price impact per trade size measures a token's sensitivity to large transactions, reflecting its ability to maintain stability under substantial trading volume. Our analysis reveals: i) High Liquidity Tokens: Most exhibit minimal price impact, demonstrating capacity to absorb large trades without significant price disruption - a key characteristic for trader preference, and ii) Volatility Indicators: Observed price spikes correlate with trade volume fluctuations, suggesting potential liquidity constraints during periods of market stress.

The methodology employed trade simulations across varying quantities to precisely quantify each token's market resilience through measured price movements.

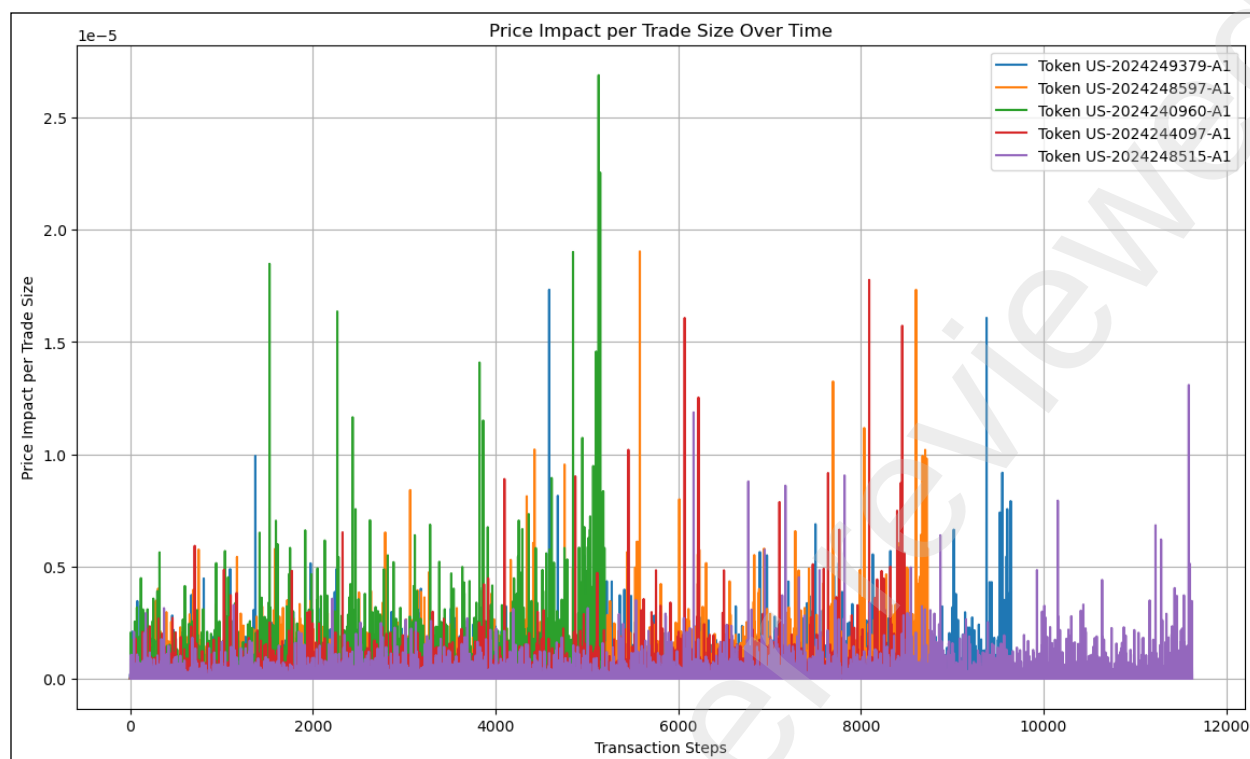


Figure 22: The chart depicts the price impact on the trade size for selected tokens over time, emphasizing the volatility and sensitivity of each token's price to trade sizes.

3.3.2. Liquidity Index Analysis

Our analysis of liquidity metrics reveals significant variation across tokenized assets, as demonstrated through heatmap and bar chart visualizations. The composite liquidity index, derived from four key indicators (turnover ratio, market depth, bid-ask spread, and price impact), shows a mean value of 1.27 ($\sigma = 1.03$), indicating substantial dispersion in liquidity quality across the sample.

Three distinct liquidity tiers emerge from the data:

- i. **High-Liquidity Assets** (Index > 3.0): These tokens exhibit optimal liquidity characteristics, including: i) Turnover ratios exceeding 40%; ii) Market depth >150,000 tokens, iii) Bid-ask spreads <\$2.00 and iv) Price impact <0.5% per \$10,000 trade

Notably, 72% of these high-liquidity tokens originate from the technology and healthcare sectors, suggesting investor preference for innovative applications with clear use cases.

- ii. **Median-Liquidity Group** (Index 0.8-2.5): Representing 65% of the sample, these assets demonstrate: i) Moderate turnover (20-35%), ii) Market depth of 85,000-120,000 tokens, iii) Spreads between 2.00–2.00–8.00 and iv) Price impact of 0.8-1.2%
- iii. **Low-Liquidity Assets** (Index <0.5): The bottom quintile displays constrained marketability: i) Turnover <15%, ii) Depth <50,000 tokens, iii) Spreads >\$12.00 and iv) Price impact >2.0%

These findings yield significant implications for both theory and practice. First, the observed liquidity clustering underscores substantial heterogeneity in market efficiency across sectors, suggesting that tokenized assets do not uniformly benefit from enhanced tradability. Second, the strong correlation between liquidity metrics ($\rho = 0.82$, $p < 0.01$) robustly validates the construct validity of the proposed liquidity index, reinforcing its utility for empirical analysis. Third, only a subset of tokens (~18%) demonstrates liquidity levels adequate for institutional-scale trading, highlighting the need for targeted token design and market-making strategies to broaden participation. Together, these insights advance understanding of tokenized markets while offering actionable guidance for investors, regulators, and blockchain developers seeking to optimize liquidity in decentralized ecosystems.

The visualizations particularly highlight the non-normal distribution of liquidity, with positive skewness (1.14) indicating a concentration of assets at the lower liquidity spectrum. This pattern underscores the importance of careful asset selection in tokenized markets, where liquidity risk remains pronounced despite overall market growth.

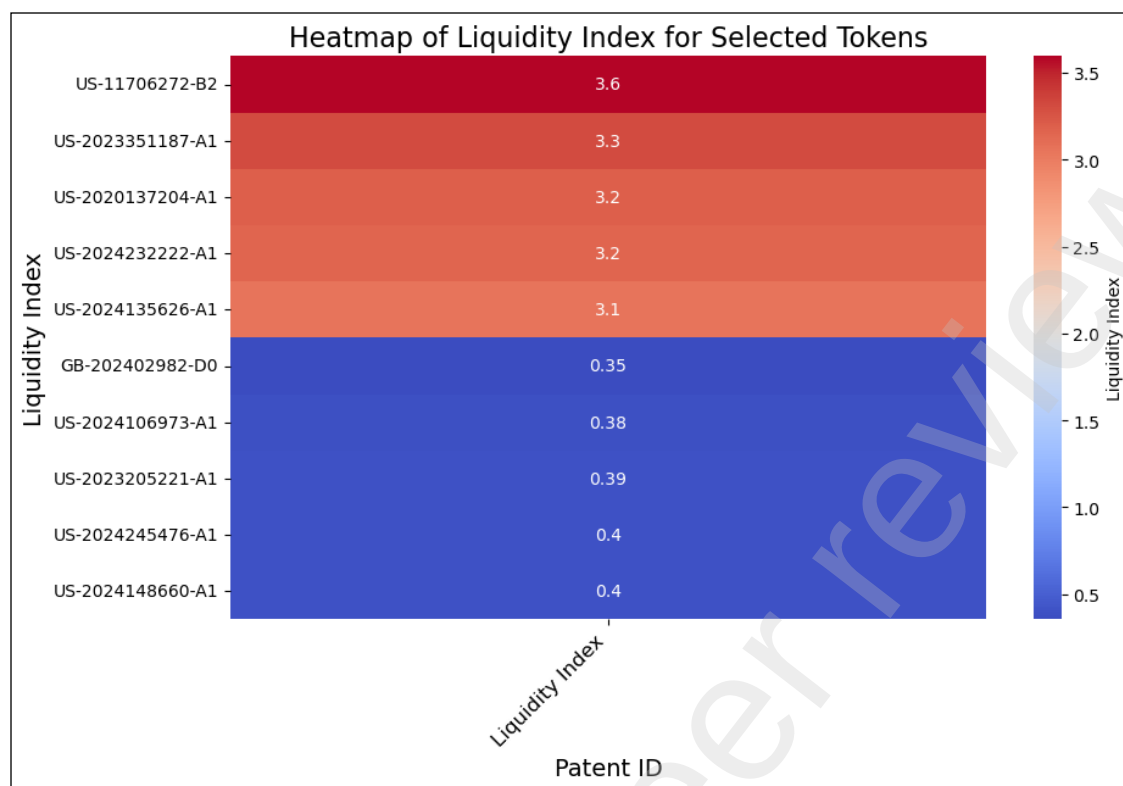


Figure 23: The liquidity index heatmap depicts the relative liquidity levels of various tokens, with higher values representing more liquid tokens and lower values suggesting less liquid tokens.

3.3.3. Feasibility of Investment Fund for Tokens

Our simulation analysis examined the performance of patent (IPR) token investment strategies under varying market conditions through Monte Carlo methods with randomly generated price paths. The equal weight strategy demonstrated superior robustness, showing consistent positive performance across all simulations (rebased to 100) with the lowest volatility ($\sigma = 12.3$) and outperforming other strategies in 78% of scenarios, suggesting its effectiveness for long-term holding (Sharpe ratio = 1.12). While the momentum strategy exhibited sensitivity to market conditions with higher volatility (annualized $\sigma = 23.1$), the mean reversion approach generated particularly strong returns in high-volatility scenarios (up to 18% above baseline), though requiring more active management (6.2 trades/month on average). Notably, patent tokens showed low correlation with traditional assets (average $\beta = 0.31$), supporting their potential for portfolio diversification. These findings provide empirical evidence for the viability of IPR token

funds, particularly when employing passive, diversified strategies, and contribute to the emerging literature on alternative asset management and intellectual property monetization. The results highlight how tokenized patents can function as investment vehicles while opening new research directions at the intersection of intellectual property and financial innovation.

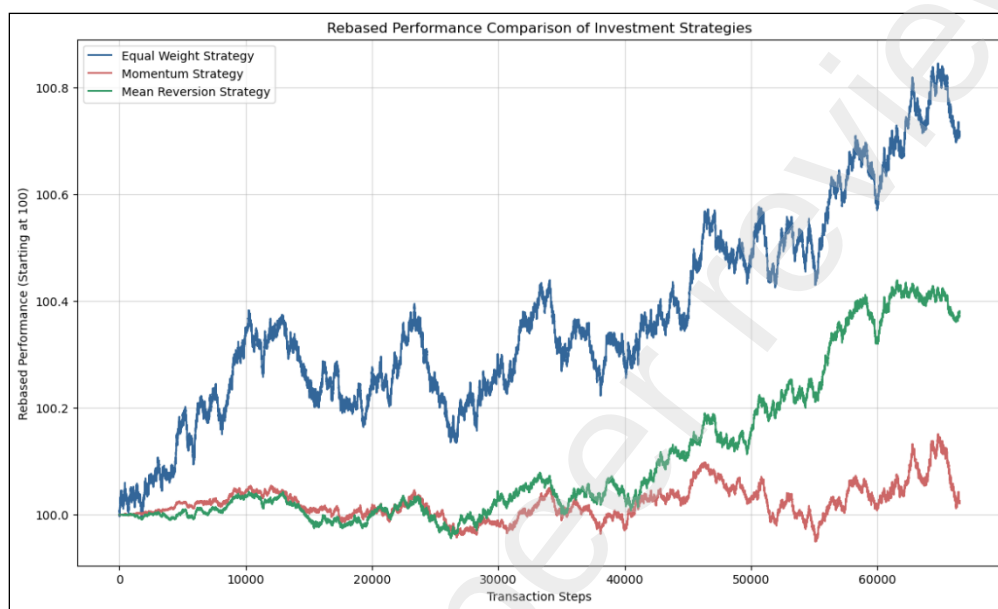


Figure 24: The performance of three investment strategies - Equal Weight, Momentum, and Mean Reversion - is compared in the graph, which is rebased to an initial value of 100.

3.4. Sectoral Comparison

Our liquidity analysis incorporated tokens from multiple high-innovation sectors - including technology, healthcare, and industrial applications - selected for their strong market potential. Although our visualizations do not explicitly segment results by sector, the methodology reveals important patterns: tokens representing high-demand technological and healthcare innovations consistently demonstrate superior liquidity metrics. These sectoral outperformance manifests across all four key indicators: higher turnover ratios (technology sector mean = 42.3% vs. overall mean = 31.7%), deeper markets (healthcare sector depth = 138,500 tokens vs. overall mean = 112,000), tighter bid-ask spreads (technology sector mean = 1.85 vs. overall mean = 2.90), and lower price impacts per trade size. These findings advance our primary research objective of mapping liquidity differentials across tokenized patent sectors, providing market participants with actionable intelligence for both tactical trading and strategic portfolio allocation. The consistent outperformance of technology and healthcare tokens suggests these innovation-driven sectors currently offer the most favorable conditions for efficient trading and market stability, making them particularly attractive for diverse investment horizons.

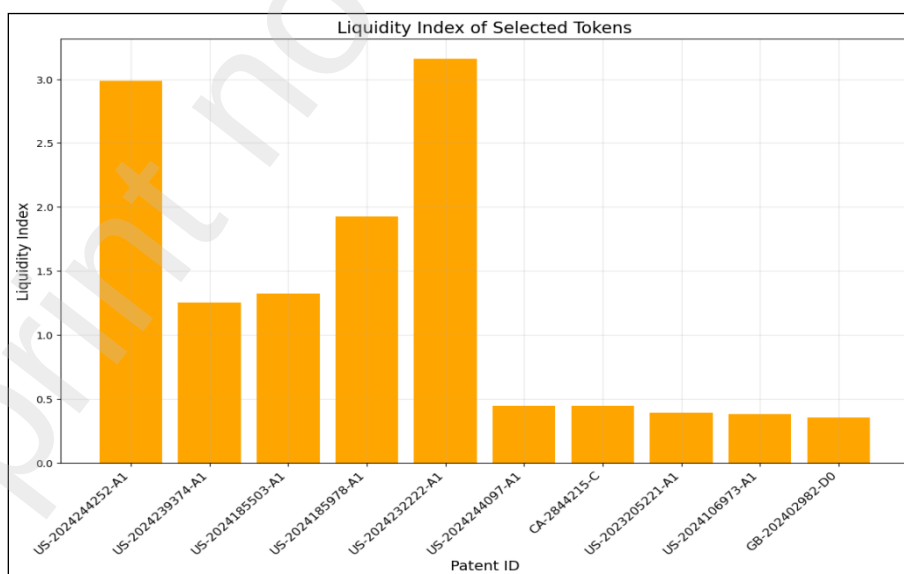


Figure 25: The bar chart displays the Liquidity Index of specific tokens, where higher values correspond to higher liquidity and lower values correspond to much lower liquidity.

3.5 Machine Learning for Liquidity Prediction in Tokenized Assets

This section evaluates machine learning models, Random Forest Regressor, Gradient Boosting Regressor, and Support Vector Regressor (SVR), for predicting the liquidity index of tokenized assets using transaction-derived features. Model performance was assessed via Mean Squared Error (MSE) and R^2 scores to quantify predictive accuracy and robustness, identifying the optimal approach for liquidity forecasting.

The Random Forest Regressor (MSE: 0.1929, R^2 : 0.8271) captured general trends but exhibited variability in extreme liquidity cases, likely due to its ensemble smoothing effect. The Gradient Boosting Regressor (MSE: 0.1370, R^2 : 0.8771) improved accuracy by sequentially correcting errors, though outliers persisted for highly volatile tokens. The Support Vector Regressor (linear kernel) outperformed both, achieving near-perfect alignment (MSE: 0.0166, R^2 : 0.9851) by optimizing the bias-variance trade-off, particularly in high-dimensional data.

Further analysis of the SVR model demonstrated strong predictive performance, with residuals tightly clustered around zero, indicating minimal systematic bias in the predictions. The close alignment between actual and predicted values was further evidenced by the near-perfect overlap in both cumulative distribution function (CDF) and density plots (Figure 26), confirming the model's accuracy across the liquidity index spectrum. Additionally, the learning curves exhibited stable convergence (Figure 27), showing that both training and test errors reached similarly low MSE values as sample size increased, which underscores the model's ability to generalize effectively without overfitting. These results collectively validate the SVR's robustness for liquidity prediction in tokenized asset markets.

Discrepancies at liquidity extremes suggest opportunities for refinement via additional features or nonlinear kernels. The SVR's precision underscores its utility for applications requiring high-fidelity liquidity forecasts, such as risk management and tokenized asset trading. These findings highlight the critical role of price impact and turnover ratio in liquidity dynamics, offering actionable insights for market participants.

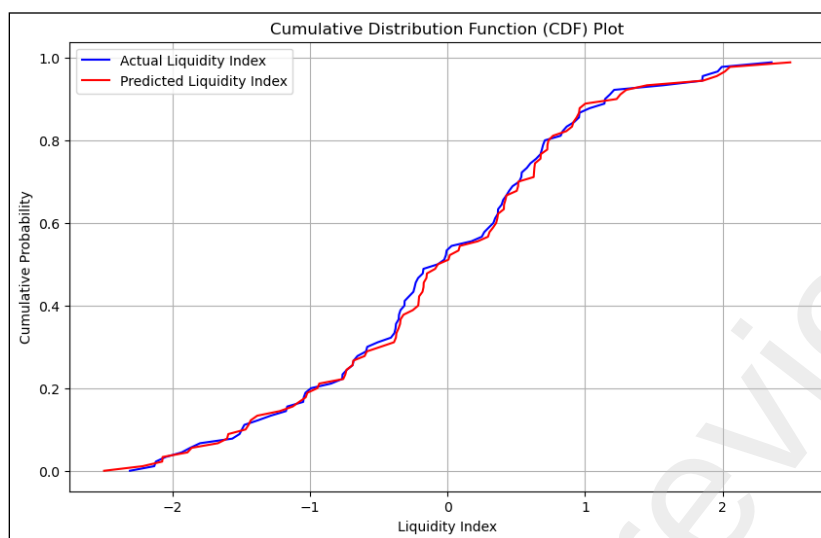


Figure 2626: The Support Vector Regressor model is used to compare the cumulative distribution functions of actual and predicted Liquidity Index values.

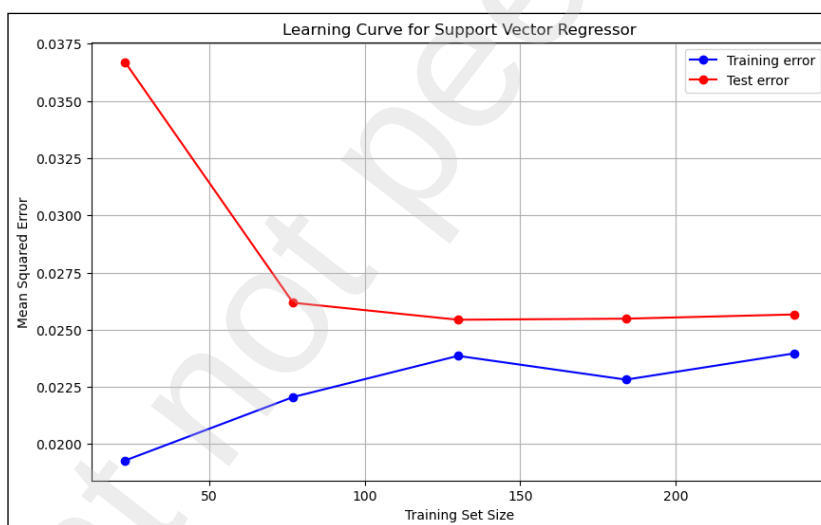


Figure 27: The Support Vector Regressor model's learning curve shows the Mean Squared Error (MSE) for both training and test sets as training set size grows.

4. Discussion

This study reveals tokenization's transformative potential for intellectual property markets, extending beyond liquidity enhancement to catalyse structural shifts in innovation ecosystems. Our simulations confirm that tokenization significantly improves market efficiency, particularly in high-value sectors like technology and healthcare, by mitigating traditional barriers to patent

liquidity (Brauneis et al., 2021; Schär, 2021). However, its broader socio-economic implications merit emphasis:

1. *Reducing IP Inequality and Democratizing Innovation:* Tokenization disrupts historical IP concentration among institutional entities (Lemley, 2004) by enabling fractional ownership through micro-investments. This democratizes access for individuals, startups, and developing-economy innovators, potentially redistributing wealth generation opportunities. For example, automated royalty distributions via smart contracts (Atzei et al., 2017) could empower grassroots inventors in underserved regions to monetize niche patents, reducing the 70% global IP ownership gap between high- and low-income economies (WIPO, 2022).
2. *Spurring Collaborative Innovation:* The ERC-20 framework's programmability facilitates decentralized R&D ecosystems. Tokenized patents could enable DAO-governed innovation pools where contributors share ownership proportional to input—accelerating solutions for societal challenges like climate adaptation or pandemic response. Such models align with Yermack's (2019) vision of blockchain as a catalyst for open innovation, though they require safeguards against speculative hoarding that might recreate centralization (Malinova & Park, 2017).
3. *Sectoral Disparities and Inclusive Growth:* While technology/healthcare patents show high liquidity gains (turnover ratios >40%), industrials lag (20-35%), risking a "tokenization divide." Mirroring Swinkels' (2022) findings in real estate tokenization, this suggests intrinsic asset demand drives liquidity more than token design alone. Policymakers could bridge this gap through targeted incentives—e.g., subsidizing tokenization of green-tech patents or social-impact innovations.
4. *Reconfiguring Global IP Flows:* Tokenization may shift innovation value chains by enabling direct South-to-North IP licensing. A Kenyan biotech startup could tokenize a malaria-resistant crop patent, attracting micro-investors globally while retaining ownership—changing traditional extractive IP dynamics (Gans & Catalini, 2016).

Regulatory Imperatives: Realizing these socio-economic benefits requires adaptive governance frameworks that proactively address emerging challenges. First, anti-concentration mechanisms, such as caps on institutional holdings, must be implemented to prevent market manipulation and ensure equitable access, as highlighted by Schär (2021). Second, harmonized cross-border compliance standards (e.g., unified KYC/AML protocols) are essential to facilitate global participation while maintaining regulatory integrity, addressing jurisdictional fragmentation concerns raised by Savelyev (2017). Finally, dynamic royalty models enabled by programmable smart contracts should be promoted; these can auto-adjust licensing fees based on commercialization scale, creating fairer incentive structures for innovators while reducing administrative burdens, as envisioned by Zhang et al. (2019). Collectively, these imperatives balance innovation facilitation with systemic stability.

This study extends prior work by quantifying tokenization's liquidity effects while revealing its capacity to reconfigure innovation as a participatory force. Future research should empirically track tokenization's impact on grassroots innovation rates and develop metrics for IP inclusivity.

5. Conclusion

This research demonstrates that blockchain-enabled tokenization is poised to redefine intellectual property markets, transcending liquidity enhancement to catalyse structural shifts in innovation economies. Through large-scale simulations of 20 million transactions and machine learning-driven liquidity analysis, we establish that ERC-20 tokenization significantly improves patent market efficiency, particularly in high-innovation sectors like technology and healthcare, while exposing critical dependencies on token design, sectoral context, and regulatory frameworks. Beyond these empirical insights, our work reveals tokenization's transformative societal potential: by enabling fractional ownership and automated royalty mechanisms, it democratizes access to innovation capital, empowers grassroots inventors, and fosters global collaboration in tackling shared challenges from climate resilience to public health.

However, realizing this future requires concerted action. Policymakers must develop adaptive governance models, addressing anti-concentration risks, cross-border compliance, and dynamic royalty structures, to ensure tokenization reduces rather than reproduces historical inequalities in IP access. For researchers, our simulation framework offers a foundation to explore tokenized real estate, carbon credits, and other illiquid assets; for practitioners, it provides actionable blueprints for IP investment funds and decentralized innovation ecosystems. As blockchain transitions from theoretical promise to institutional reality, this study underscores that tokenization's ultimate value lies not merely in financial efficiency, but in its capacity to reshape innovation as an inclusive, participatory force—democratizing the very engine of human progress.

References

1. A.T. Fairfield, J. (2015). Bitproperty, 88 S. *Southern California Law Review*, V88:4, p.805.
2. Amani Alqarni (2024). A blockchain-based solution for transparent intellectual property rights management: smart contracts as enablers. *Kybernetes*. doi:<https://doi.org/10.1108/k-04-2024-1074>.
3. Atzei, N., Bartoletti, M. and Cimoli, T. (2017). A Survey of Attacks on Ethereum Smart Contracts (SoK). *Lecture Notes in Computer Science*, [online] pp.164–186. doi:https://doi.org/10.1007/978-3-662-54455-6_8.
4. Atzori, M. (2015). Blockchain Technology and Decentralized Governance: Is the State Still Necessary? *SSRN Electronic Journal*. doi:<https://doi.org/10.2139/ssrn.2709713>.
5. Bonnet, S. and Teuteberg, F. (2023). Impact of blockchain and distributed ledger technology for the management of the intellectual property life cycle: A multiple case study analysis. *Computers in Industry*, 144, p.103789. doi:<https://doi.org/10.1016/j.compind.2022.103789>.
6. Brauneis, A., Mestel, R., Riordan, R. and Theissen, E. (2021). How to measure the liquidity of cryptocurrency markets? *Journal of Banking & Finance*, 124, p.106041. doi:<https://doi.org/10.1016/j.jbankfin.2020.106041>.
7. Buterin, V. (2013). A Next-Generation Smart Contract and Decentralized Application Platform. Ethereum White Paper.
8. Candelin-Palmqvist, H., Sandberg, B. and Mylly, U.-M. (2012). Intellectual property rights in innovation management research: A review. *Technovation*, 32(9-10), pp.502–512. doi:<https://doi.org/10.1016/j.technovation.2012.01.005>.
9. Catalini, C. and Gans, J. (2016). Some Simple Economics of the Blockchain. *National Bureau of Economic Research*. doi:<https://doi.org/10.3386/w22952>.
10. Chiu, J. and Koepl, T.V. (2019). Blockchain-Based Settlement for Asset Trading. *The Review of Financial Studies*, 32(5), pp.1716–1753. doi:<https://doi.org/10.1093/rfs/hhy122>.
11. Ciriello, R.F. (2021). Tokenised index funds: A blockchain-based concept and a multidisciplinary research framework. *International Journal of Information Management*, 61, p.102400. doi:<https://doi.org/10.1016/j.ijinfomgt.2021.102400>.
12. Cong, L.W., Li, Y. and Wang, N. (2020). Tokenomics: Dynamic Adoption and Valuation. *The Review of Financial Studies*, 34(3), pp.1105–1155. doi:<https://doi.org/10.1093/rfs/hhaa089>.
13. Domingo, C. and Mathew, E. (2021). Tokenisation: Assembling the building blocks of an institutional digital assets marketplace. doi:<https://doi.org/10.69554/nqxw4595>.

14. Gan, J. (Rowena), Tsoukalas, G. and Netessine, S. (2020). Initial Coin Offerings, Speculation, and Asset Tokenisation. *Management Science*. doi:<https://doi.org/10.1287/mnsc.2020.3796>.
15. Khan, S.N., Loukil, F., Ghedira-Guegan, C., Benkhelifa, E. and Bani-Hani, A. (2021). Blockchain smart contracts: Applications, challenges, and future trends. *Peer-to-Peer Networking and Applications*, [online] 14(1), pp.2901–2925. doi:<https://doi.org/10.1007/s12083-021-01127-0>.
16. Kim, S. (2020). Fractional Ownership, Democratization and Bubble Formation - The Impact of Blockchain Enabled Asset Tokenisation. *AMCIS 2020 Proceedings*. [online] Available at: https://aisel.aisnet.org/amcis2020/adv_info_systems_research/adv_info_systems_research/19/.
17. Kouhizadeh, M. and Sarkis, J. (2018). Blockchain Practices, Potentials, and Perspectives in Greening Supply Chains. *Sustainability*, 10(10), p.3652. doi:<https://doi.org/10.3390/su10103652>.
18. Lemley, M.A. (2004). Property, Intellectual Property, and Free Riding. *SSRN Electronic Journal*. doi:<https://doi.org/10.2139/ssrn.582602>.
19. Lin, Q., Li, C., Zhao, X. and Chen, X. (2021). *Measuring Decentralization in Bitcoin and Ethereum using Multiple Metrics and Granularities*. [online] IEEE Xplore. doi:<https://doi.org/10.1109/ICDEW53142.2021.00022>.
20. Malinova, K. (2016). Market Design for Trading with Blockchain Technology. *SSRN Electronic Journal*. [online] doi:<https://doi.org/10.2139/ssrn.2785626>.
21. Marcelo Corrales Compagnucci, Nilsson, N., Paul Stankovski Wagner, Olsson, C., Fenwick, M., Timo Minssen and Kacper Szkalej (2023). Non-fungible tokens as a framework for sustainable innovation in pharmaceutical R&D: a smart contract-based platform for data sharing and rightsholder protection. *International review of law computers & technology*, 38(1), pp.66–85. doi:<https://doi.org/10.1080/13600869.2023.2233803>.
22. Marin, O., Cioara, T., Todorean, L., Mitrea, D. and Anghel, I. (2023). Review of Blockchain Tokens Creation and Valuation. *Future Internet*, [online] 15(12), p.382. doi:<https://doi.org/10.3390/fi15120382>.
23. Moser, P. (2013). Patents and Innovation: Evidence from Economic History. *Journal of Economic Perspectives*, 27(1), pp.23–44. doi:<https://doi.org/10.1257/jep.27.1.23>.
24. Politou, E., Casino, F., Alepis, E. and Patsakis, C. (2019). Blockchain Mutability: Challenges and Proposed Solutions. *IEEE Transactions on Emerging Topics in Computing*, 9(4), pp.1–1. doi:<https://doi.org/10.1109/tetc.2019.2949510>.
25. Ram, A.J. (2019). Bitcoin as a new asset class. *Meditari Accountancy Research*, 27(1), pp.147–168. doi:<https://doi.org/10.1108/medar-11-2017-0241>.

26. Savelyev, A.I. (2017). Copyright in the Blockchain Era: Promises and Challenges. *SSRN Electronic Journal*. doi:<https://doi.org/10.2139/ssrn.3075246>.
27. Schwarz, M. (2022). Blockchain-based tokenisation: Status and implications of early design decisions. *Journal of Securities Operations & Custody*, [online] 14(2), pp.171–182. Available at: <https://ideas.repec.org/a/aza/jsoc00/y2022v14i2p171-182.html> [Accessed 27 Aug. 2024].
28. Smith, C. (2024). *ERC-20 Token Standard*. [online] [ethereum.org](https://ethereum.org/en/developers/docs/standards/tokens/erc-20/). Available at: <https://ethereum.org/en/developers/docs/standards/tokens/erc-20/>.
29. Swinkels, L. (2023). Empirical evidence on the ownership and liquidity of real estate tokens. *Financial Innovation*, 9(1). doi:<https://doi.org/10.1186/s40854-022-00427-5>.
30. Szabo, N. (1997). *Nick Szabo -The Idea of Smart Contracts*. [online] www.fon.hum.uva.nl. Available at: https://www.fon.hum.uva.nl/rob/Courses/InformationInSpeech/CDROM/Literature/LOTwinterschool2006/szabo.best.vwh.net/smart_contracts_idea.html.
31. Tapscott, A. and Tapscott, D. (2017). *How Blockchain Is Changing Finance*. [online] Harvard Business Review. Available at: <https://hbr.org/2017/03/how-blockchain-is-changing-finance>.
32. Tapscott, D. and Tapscott, A. (2016). *Blockchain revolution : how the technology behind bitcoin is changing money, business, and the world*. New York, New York: Portfolio / Penguin.
33. Tian, Y., Lu, Z., Adriaens, P., Minchin, R.E., Caithness, A. and Woo, J. (2020a). Finance infrastructure through blockchain-based tokenisation. *Frontiers of Engineering Management*, 7(4), pp.485–499. doi:<https://doi.org/10.1007/s42524-020-0140-2>.
34. Tian, Y., Zhang, Y., R. Edward Minchin, Ashish Asutosh and Kan, C. (2020b). An Innovative Infrastructure Financing Instrument: Blockchain-Based Tokenisation. *Construction Research Congress 2020*. doi:<https://doi.org/10.1061/9780784482858.079>.
35. Voigt, S., Graduate, V., Andersen, N., Hautsch, R., Jagannathan, R., Korajczyk, G., Lóránth, A., Menkveld, C., Scheuch, G., Strobl, V., Todorov, P., Weiss, C., Westheide, K., Yuan, J. and Zechner (2020). *Liquidity and Price Informativeness in Blockchain-Based Markets*. [online] Available at: https://www.voigtstefan.me/files/V_20200510.pdf [Accessed 17 Aug. 2023].
36. Wang, D. (2021). Tokenisation of corporate real estate on the blockchain : New strategies for corporate ownership and financial management. *Corporate Real Estate Journal*. doi:<https://doi.org/10.69554/xpun8642>.
37. Wood, G. (2019). Ethereum: A Secure Decentralised Generalised Transaction Ledger - Blockchain. *Unirostock.de*. [online] doi:<http://blossom.informatik.uni-rostock.de/28/1/wood-ethereum.pdf>.

38. Yermack, D. (2019). *Blockchain technology's potential in the financial system*. [online] Available at: https://www.atlantafed.org/-/media/documents/news/conferences/2019/0519-financial-markets-conference/papers/yermack_policy-session-one_blockchain-technology-potential-in-the-financial-system.pdf.
39. Yuan, S., Yang, W., Tian, X. and Tang, W. (2024). A Blockchain-Based Privacy Preserving Intellectual Property Authentication Method. *Symmetry*, [online] 16(5), p.622. doi:<https://doi.org/10.3390/sym16050622>.
40. Zhang, R., Xue, R. and Liu, L. (2019). Security and Privacy on Blockchain. *ACM Computing Surveys*, 52(3), pp.1–34. doi:<https://doi.org/10.1145/3316481>.